Gerard Bosch

Software Engineer,
Freelancer/Contractor

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"The purpose of abstraction is not to be vague, but to create a new semantic level in which one can be absolutely precise."
—Edsger W. Dijkstra

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[*Read the one-page résumé version at* resume.gerardbosch.xyz]

Summary

- Enthusiast about Functional Programming, software engineering/craftsmanship, clean code, architectural/design patterns and programming languages theory.
- Good at identifying inefficiencies, finding solutions and improving processes. I enjoy collaborating with diverse teams to solve challenging problems.

My background is mostly about designing/implementing backend systems and APIs. I have good analysis skills and can break down requirements well and transform that into code.

Really concerned about the code itself, its readability, expressiveness, precision and conciseness. Continuous improvement advocate & continuous refactor and code reuse as a mantra. Accidental-complexity fighter. Curious by nature, proactive and self-taught. Like learning 😃, love crafting ❤️.

Presentations

2020 Functional Programming super-short intro
2018 A technical introduction to Blockchain technology

Professional Experience

2025 **Software Engineer | Messaging + Recommendation**, *Leboncoin.fr* · *Freelance*, Remote

🎯 *Support the migration and handover of Adevinta's central messaging (chat) backend platform to Leboncoin, ensuring smooth transition to full ownership and autonomy. The system supports 20M+ monthly active users, tens of millions of conversations per month, and millions of daily messages through an Event-Driven Architecture (Kafka, Kafka Streams).*

🎯 *Created an architectural framework (Spring based) to be an umbrella of 30+ microservices and centralize service management, versioning and evolution, providing cross-cutting components for the Event-Driven system, and make maintenance and upgrades much easier. It strongly contributed to harmonize code.*

- Actively engage in the roadmap for 2026 through actionable initiatives.
- Develop new features and modernize the codebase and microservices architecture to meet evolving business and system requirements, improving scalability, reliability, and maintainability.
- Operate, troubleshoot, and optimize the system to ensure high availability, performance, and resilience in a large-scale distributed environment. Bridge original and new teams, driving knowledge transfer, collaboration, and alignment for successful migration.

After the first year I switched to the Recommendation team to work on the Golang Backend for the Homepage & Recommendation team which serves high-volume traffic and provides user personalized recommendations; interfacing with Frontend, ML and vector DB for ANN search and semantic search.

Keywords: *Kotlin, Spring Boot, Kafka, Kafka Streams, Kubernetes, ScyllaDB, Postgres, Clickhouse, AWS, Go, Milvus/Zilliz...*

2024–2025 **Backend Software Engineer | Quarkus, Reactive**, *Tymit*, Remote

🎯 *Authored an RFC with actionable proposals to align the current architecture with Clean Architecture™ & SOLID principles, with a special focus on the Interface Segregation (ISP) and Single Responsibility Principle (SRP), plus other enhancements.*

- Engaged with the integrations team to support the migration of hundreds of thousands of customers into our system, while also developing new public-facing APIs to enable a new business model requiring first-party and third-party B2B backend integrations.
- Improved testing reliability by introducing Property-Based Testing and showcasing its principles to the backend guild, ensuring more robust testing including edge-cases, generating thousands of test values.
- Spearheaded API contract segregation into dedicated modules, enabling the publication of versioned API artifacts for seamless and efficient cross-service integration using Gradle, contributing to DX.
- Shared coding best practices and actively contributed to code reviews and BE discussions.

Keywords: *Quarkus, Microservices, Reactive Programming, AWS (ECS, DynamoDB, SQS, SNS, RDS), Liquibase,...*

2022–2023 **(Break) / Software Engineer**, *Self employed*, Barcelona

🎯 *I set up my own company to provide software engineering services as a contractor.*

- Algorithmic trading, research, design & implementation of a risk management system for loss control, as well as key metrics collection/reporting using DDD & Hexagonal arch.
- High level design of a multi-level affiliation system with referral fee distribution.
- Took a break from industry to learn about finances, markets and investments.

2021–2022 **Software Engineer**, *N26 Bank*, Barcelona

🎯 *I contributed to streamline the processing of banking fees with the goal of having a more centralized and well-defined domain for charging fees.*

Joined the fees team, which is in charge of designing and operating a new fee platform intended to be the new-standardized way of charging company-wide banking fees.

- That implied not only the creation of a dedicated microservice, also re-purposing and refactoring of existent ones along with APIs, as well as DB migrations.
- Implement new business use cases as per new fee creation or regulatory compliance.
- Being a bank, testing is one of the most important and cared matters (pyramid testing, contract testing, regression, TDD,...).

Keywords: *Kotlin, Spring Boot, Microservices, EDA, Kafka, REST, Postgres, Datadog, ELK, Kibana, AWS, Hexagonal, Clean Architecture, Domain Driven Design.*

2016–2021 **Senior Software Engineer**, *GFT IT Consulting*, Lleida/Barcelona

🎯 *Successfully delivered a large and complex regulatory banking project to provide external, permission-based access to banking APIs, ensuring on-time completion. Then, I was called to contribute building a new bank architecture for high-concurrency, based on acquired know-how.*

PROJECT HIGHLIGHTS (most relevant projects only):

- *Banc Sabadell*: Middleware Architecture (2020)
  Role: Software Engineer
  Start redefining from scratch a whole new framework and development environment based on MsA, Spring Boot, API First, Event Driven, DDD, Hexagonal and so forth.
- *Banc Sabadell*: PSD2 & APIfication (2018–2020)
  Role: Tech Lead (development team)
  • Design and implement the APIs for the European regulatory project *Payments Service Directive 2*/PSD2 (*Strong Customer Authentication*/SCA, Open Banking, Fintech integration,...); leading the first approach to API exposition for the *Banc Sabadell*. I worked both on the API definition side and the Architecture side, defining and implementing architectural components for the Microservices Architecture, where I was established as the technical lead for the development team until post-production.
  • Defined and set up an *API-First* approach with OpenAPI and API contracts generation (interfaces + DTO) with CI integration, making API definition agnostic and decoupled from its implementation.
- *Bankinter*: Microservice architecture (2017)
  Role: Developer
  Contribute in the development of an architecture for Bankinter based on MsA (Micro Services Architecture) with Java 8 and Spring Boot/Cloud. The built Architecture provides the pre-configured components: from security, logging, audit,... to cryptography; to allow developers to easily get started implementing microservices with not much Spring background required.

Keywords: *Spring, Java, Vavr, FP, OpenShift, ELK, Kibana, OpenAPI, API-First, Blockchain.*

2014–2016 **Software Engineer**, *ICG Software*, Torrefarrera, Lleida

Senior developer in charge of mobile applications for retail.

- Port a whole Android retail selling application to Windows Phone platform due to customer demands. In the way, learn both Android and WP frameworks.
- Build from scratch a new application for warehouse and inventory management; featuring integrated in-app barcode scanner through phone's camera, Android object database, and dual online/offline working mode.
- Research and use of cutting-edge technologies and libraries (Android data binding, Realm databases, Gson parser, Retrofit...).
- Architectural and design patterns implementation.

Keywords: Java, C#/.NET, Mobile apps, Android, Windows Phone, Realm, REST.

2011–2013 **Researcher/developer**, *Artificial Intelligence Research Institute (IIIA)-Spanish National Research Council (CSIC)*, Bellaterra

Hired for the research project Innpacto-2011 (Spanish Science and Innovation Ministry), *NEWMATICA: Intelligent and Energy Efficient Advanced System for Vacuum Waste Collection* (IPT-2011-1496-310000). In collaboration with University of Lleida.

- Research on artificial intelligence algorithms and machine learning.
- Design and implementation of a simulator, modelling of real plants topology and operation. Experimentation running bulk simulations in a cluster and automatic processing and representation of results.
- Implementation of algorithms based on *Approximate Dynamic Programming* (Reinforcement Learning) in order to optimize operational plans of waste collecting plants.

Keywords: Approximate Dynamic Programming, ADP, Discrete Event Simulation, DES, Integer Linear Programming, ILP, modelling, data collection, AI, research, planning, optimization, Python.

2010–2011 **Systems/comms operations**, *Information Systems and Communications Area, University of Lleida*, Lleida

- Setting up and deployment of Open-Exchange system (*Open Source*) covering the need of a corporate *Groupware* for the university staff. Set up, deployment, integration and maintenance of the service. Integration of the system with the corporate mobile phones.
- Setting up of an open source system for the management of name servers (DNS) in the university.
- Installation and management of GNU/Linux servers and virtual machines.

Keywords: Linux, virtualization, servers, DNS, networking, groupware.

Technical Skills

Programming Languages: Kotlin, Scala, Java, Bash, Haskell, Python, Perl, SQL.

Interest Areas: Functional Programming, Unison lang, Effect Systems, ZIO, Kyo,... Big/Fast Data & Streaming, DLTs/Blockchain & Web3, Event-Driven Architectures, Artificial Intelligence, WebAssembly.

Development Methodologies: FP patterns, GoF design patterns, SOLID, Clean Code, Hexagonal/Clean Architecture, 12-Factor, ADTs, Programming with types, Railway Oriented Prog., Agile development, SCRUM, Domain Driven Design, TDD.

Others: Cloud-Native, AWS, Git, Maven/Gradle/sbt, JGitVer, Docker, OpenShift, Devcontainers, Testcontainers, DevOps, GitHub, GitLab, Cloudflare, LaTeX, Linux, Nix, Microservices, Cloud Native, OpenAPI, AsyncAPI, API first, oAuth.

Courses and Certifications

2020 **Big Data Foundations**, *IBM Cognitive Class*, Course badge
2020 **Hadoop Foundations**, *IBM Cognitive Class*, Course badge
2020 **Spark**, *IBM Cognitive Class*, Course badge
2020 **Big Data Foundations (level 2)**, *IBM Cognitive Class*, Course badge
2018 **Serverless Data Analysis with Google BigQuery and Cloud Dataflow**, *Coursera*, Course accomplishment
2018 **Leveraging Unstructured Data with Cloud Dataproc on Google Cloud Platform**, *Coursera*, Course accomplishment
2018 **Google Cloud Platform Big Data and Machine Learning Fundamentals**, *Coursera*, Course accomplishment

Conference Attendance

2020–2021 Several online conferences, such as **JLove, Scala Love in the City, Haskell Love,...**, *online*
2019 **API Days**, *Paris*
2018 **API Days**, *Paris*
2018 **Ethereum Community Conference**, *Paris*
2017 **JBcnConf**, *Java & JVM Conference*, Barcelona

Education

2011 **Master in Open Source Software Engineering**, *University of Lleida*, Lleida / *VIA University College* (Denmark)
2011 **IT Engineering Higher Degree**, *University of Lleida*, Lleida / *VIA University College* (Denmark)
2005–2010 **IT Engineering Bachelor**, *University of Lleida*, Lleida

Final Degree Dissertations

Master in Open Source Software Engineering

title: *Setting up and deployment of Sauron system for DNS system management at University of Lleida.*
date: 28–09–2011
advisor: Dr. Carles Mateu Piñol, Department of Computer Science and Industrial Engineering, University of Lleida, Lleida.
description: Setting up an open source system for the name servers (DNS) management of the university. Integration of several open source components together with an LDAP database into ISC BIND name server. Implementation, configuration, integration and deployment of that system in the university's production infrastructure.

***Qualified** with honors.*

📙 Dissertation » (Catalan + English full configuration guide)
🖥️ Presentation » (English)

IT Engineering Bachelor

title: *Implementation of Golay codes in SAGE.*
date: 15–01–2010
advisor: Dr. Ramiro Moreno Chiral, Department of Mathematics, University of Lleida, Lleida.
description: Implementation of a kind of channel error correcting codes (ECC) inside mathematical Open Source Software package SAGE, which is built up in Python and just previously had a generic implementation for linear codes.

***Qualified** with honors.*

📙 Dissertation »
🖥️ Presentation »

Languages

English: Professional working proficiency
B2 CEFR (Common European Framework of Reference)
Catalan: Native
Spanish: Native

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